Linear Algebra for Political Scientists

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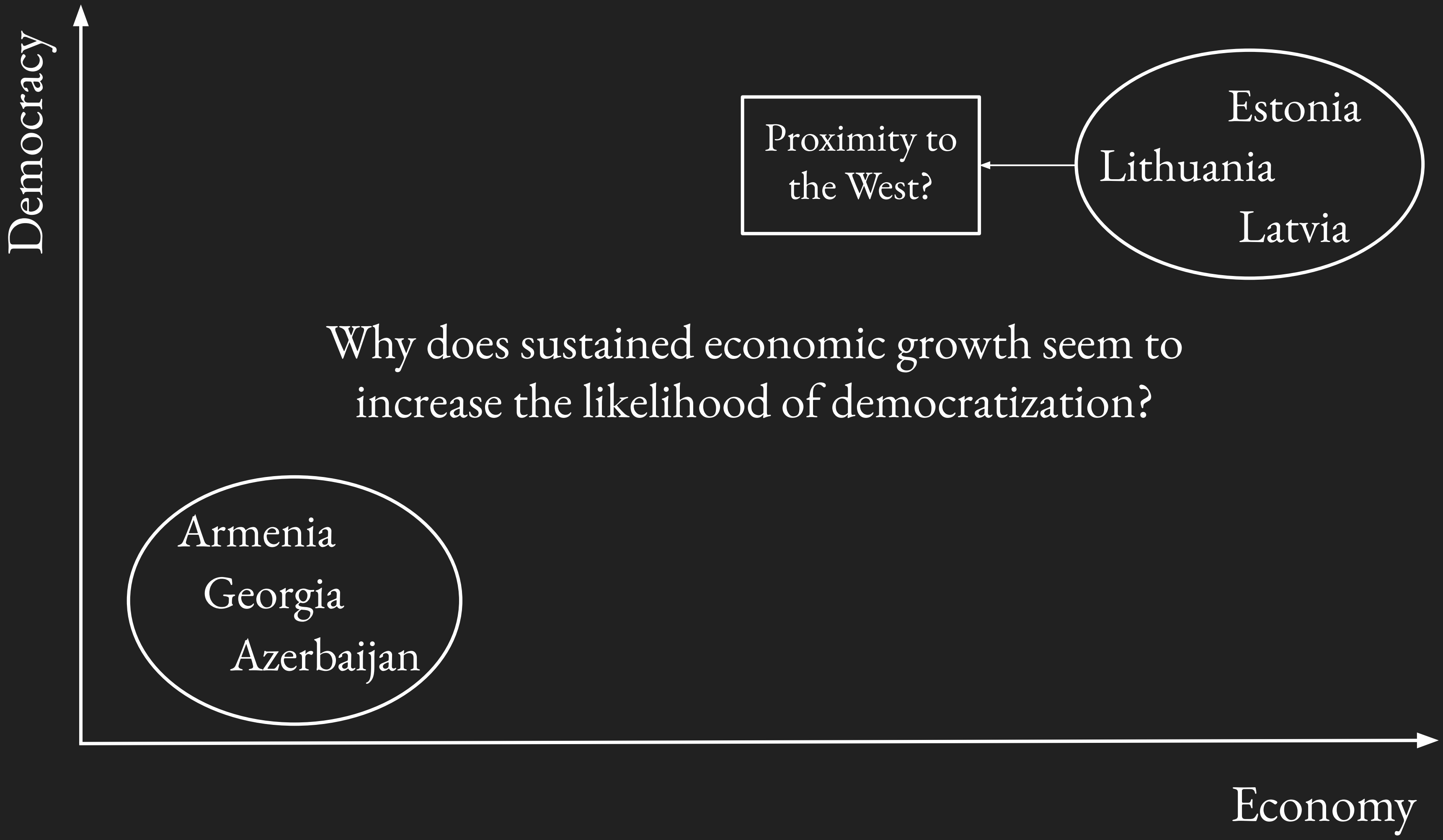
Structure

1. Motivation: Why study linear algebra?
2. Introduce the raw material (vectors, matrices)
3. Learn how to operate the heavy machinery (vector & matrix operations)


Democracy
Proximity to the West?
Estonia
Lithuania
Latvia
Why does sustained economic growth seem to increase the likelihood of democratization?
Armenia
Georgia
Azerbaijan
Economy

Country	Region	GDP Growth	EU Integration	Democracy Score
Armenia	South Caucasus	3	2	3
Azerbaijan	South Caucasus	4	1	2
Georgia	South Caucasus	3	3	4
Estonia	Baltics	7	9	8
Latvia	Baltics	6	8	7
Lithuania	Baltics	7	8	8

Matrix

Rectangular array of numbers, arranged in rows and columns.

We write matrices as uppercase letters (A), and refer to the size of a matrix as $m \times n$, meaning m rows and n columns. The entry in row i, column j is written a_{ij}

$$\begin{bmatrix} 3 & 2 & 3 \\ 4 & 1 & 2 \\ 3 & 3 & 4 \\ 7 & 9 & 8 \\ 6 & 8 & 7 \\ 7 & 8 & 8 \end{bmatrix}$$

Vector

Ordered list of numbers.

We write vectors as lowercase bold letters ($\mathbf{x}$) and their individual entries with subscripts ($x_1, \ldots, x_n$)

A vector with n entries is said to have *dimension* n, or to live in $\mathbb{R}^n$

$$\begin{bmatrix} 3 \\ 4 \\ 3 \\ 7 \\ 6 \\ 7 \end{bmatrix}$$

Note: The object to the left is a **column** vector (isolating a single column of the matrix A). We can also isolate a single row of A to obtain a **row** vector, e.g.:

$$\begin{bmatrix} 3 & 2 & 3 \end{bmatrix}$$

Vector Operations: Addition

$$\text{Let } \mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

Note: Addition is only defined when **u** and **v** have the same number of entries.

Vector Operations: Addition (Example)

$$A = \begin{array}{c} \begin{bmatrix} 3 & 2 & 3 \\ 4 & 1 & 2 \\ 3 & 3 & 4 \\ 7 & 9 & 8 \\ 6 & 8 & 7 \\ 7 & 8 & 8 \end{bmatrix} \\ \begin{array}{ccc} \text{GDP Growth} & \text{EU Integration} & \text{Democracy Score} \end{array} \end{array}$$

Call the GDP Growth and EU Integration column vectors $\mathbf{x}_1$ and $\mathbf{x}_2$, respectively, and add them together.

$$\mathbf{x}_1 + \mathbf{x}_2 = \begin{bmatrix} 5 \\ 5 \\ 6 \\ 16 \\ 14 \\ 15 \end{bmatrix}$$

Vector Operations: Addition (Example)

This is a real (if crude) *composite* score: each country's growth and integration values summed together.

Whether this sum is a meaningful quantity is a modeling question, not a mathematical one.

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Vector Operations: Scalar Multiplication

For a number c (called a *scalar* to distinguish it from a vector) and a vector $\mathbf{v}$:

$$c\mathbf{v} = \begin{bmatrix} cv_1 \\ \vdots \\ cv_n \end{bmatrix}$$

Vector Operations: Scalar Multiplication (Example)

$$A = \begin{bmatrix} 3 & 2 & 3 \\ 4 & 1 & 2 \\ 3 & 3 & 4 \\ 7 & 9 & 8 \\ 6 & 8 & 7 \\ 7 & 8 & 8 \end{bmatrix}$$

$$\begin{array}{ccc} \text{GDP Growth} & \text{EU Integration} & \text{Democracy Score} \end{array}$$

Suppose we want to report democracy score on a 0-20 scale instead of 0-10. This requires us to double each entry of the vector. Rewrite the new vector.

$$\mathbf{x}_3^* = \begin{bmatrix} 6 \\ 4 \\ 8 \\ 16 \\ 14 \\ 16 \end{bmatrix}$$

Note on Substantive Interpretation

- Political scientists use these basic vector operations all the time, whether or not we realize it
  - Vector addition allows us to create “composite scores” or indices (more on this in a moment)
  - Scalar multiplication allows us to rescale variables (e.g., convert units, standardize, normalize to a 0-1 range)
- A dual responsibility:
  - Know how to perform the mathematical operations
  - Discern whether these operations are substantively meaningful

Example: One of Many Composite Indices from V-Dem

5.10.5 Citizen-initiated component of direct popular vote index (D) (v2xdd_cic)

some composite measure
of "direct democracy"

Aggregation: This index is the normalized average of the scores of both indices of citizen-initiated mechanism of direct democracy popular initiatives and referendums. For an elaboration of the weighting factor of each component, see David Altman 2017. The index is aggregated using this formula:

$$v2xdd\_cic = [v2xdd\_i\_ci + v2xdd\_i\_rf]/4$$

Vector
Addition

Scalar Multiplication
(by 0.25)

The Dot Product

For two vectors of the same dimension, $\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$,

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Multiply corresponding entries, then add up the results.

Like addition, the dot product is only defined when both vectors have the same number of entries.

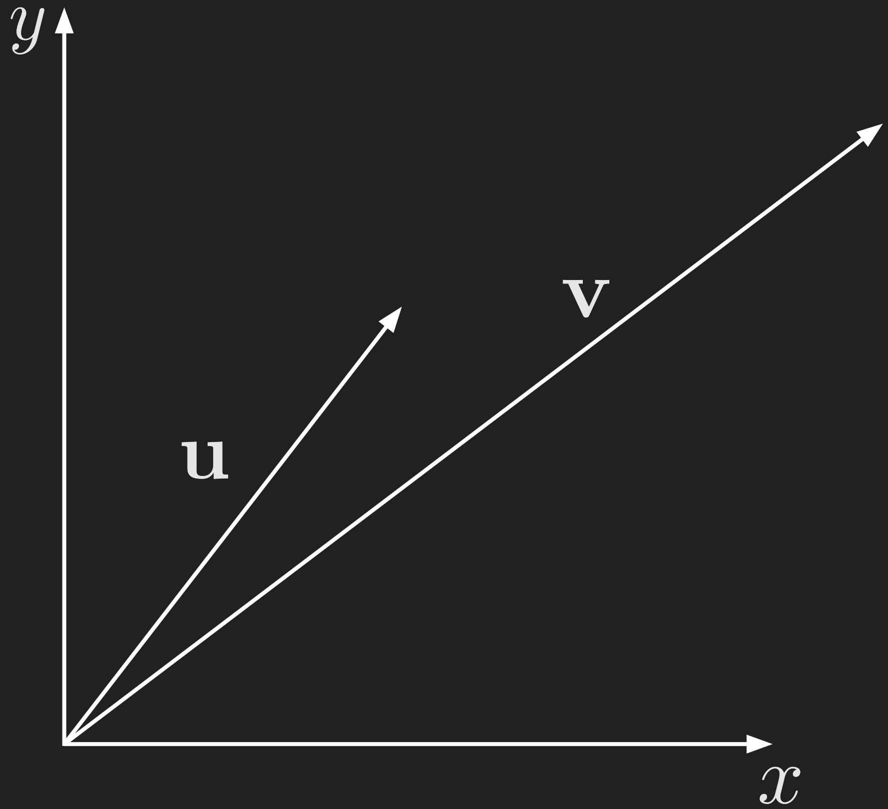
The Dot Product: Example

$$A = \begin{array}{c} \begin{bmatrix} 3 & 2 & 3 \\ 4 & 1 & 2 \\ 3 & 3 & 4 \\ 7 & 9 & 8 \\ 6 & 8 & 7 \\ 7 & 8 & 8 \end{bmatrix} \\ \begin{array}{ccc} \text{GDP Growth} & \text{EU Integration} & \text{Democracy Score} \end{array} \end{array}$$

Calculate the dot product of the GDP Growth and EU Integration columns.

$$\begin{aligned} \mathbf{x}_1 \cdot \mathbf{x}_2 &= (3)(2) + (4)(1) + (3)(3) \\ &\quad +(7)(9) + (6)(8) + (7)(8) \\ &= 6 + 4 + 9 + 63 + 48 + 56 \\ &= 186 \end{aligned}$$

Geometric Implications of the Dot Product



Beyond serving as a weighted sum, the dot product also carries information about how two vectors relate to each other in space.

To see this, we first need a way to measure a vector's own size.

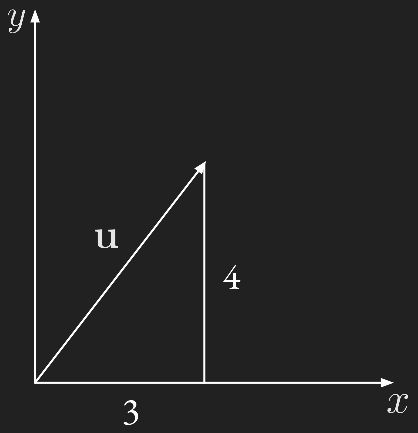
Vector Norm

The norm (length) of a vector $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Note: This is simply the dot product of a vector with itself (square-rooted). It is a direct generalization of the Pythagorean theorem to n dimensions.

$$\|\mathbf{u}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$



Dot Products and Angles

For any two vectors $\mathbf{u}, \mathbf{v}$,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between them. Equivalently,

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

The dot product measures how aligned two vectors are. $\cos \theta$ close to 1 means that the vectors point in nearly the same direction; close to 0 means they're roughly perpendicular (unrelated); close to -1 means they point in nearly opposite directions.

Dot Products and Angles: Example

$$\mathbf{x}_1 = \begin{bmatrix} 3 \\ 4 \\ 3 \\ 7 \\ 6 \\ 7 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 3 \\ 2 \\ 4 \\ 8 \\ 7 \\ 8 \end{bmatrix}$$

GDP Growth

Democracy Score

How aligned are growth and democracy?

$$\mathbf{x}_1 \cdot \mathbf{x}_2 = 9 + 8 + 12 + 56 + 42 + 56 = 183$$

$$\|\mathbf{x}_1\| = \sqrt{9 + 16 + 9 + 49 + 36 + 49}$$
$$= \sqrt{168} \approx 12.96$$

$$\|\mathbf{x}_2\| = \sqrt{9 + 4 + 16 + 64 + 49 + 64}$$
$$= \sqrt{206} \approx 14.35$$

$$\cos\theta = \frac{183}{(12.96)(14.35)} \approx 0.984 \longrightarrow$$

<u>Very</u> aligned!

Matrix-Vector Multiplication

For an $m \times n$ matrix and an $n \times 1$ vector $\mathbf{w}$, the product $M\mathbf{w}$ is the $m \times 1$ vector whose i-th entry is the dot product of M's i-th row with $\mathbf{w}$:

$$(M\mathbf{w})_i = \sum_{j=1}^{n} M_{ij} w_j$$

This is only defined when the number of columns in M matches the number of entries in $\mathbf{w}$ (each row needs a matching weight to dot against).

Matrix-Vector Multiplication: Example

$$M\mathbf{w} = \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 3 & 3 \\ 7 & 9 \\ 6 & 8 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} (3)(2) + (2)(1) \\ (4)(2) + (1)(1) \\ (3)(2) + (3)(1) \\ (7)(2) + (9)(1) \\ (6)(2) + (8)(1) \\ (7)(2) + (8)(1) \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \\ 9 \\ 23 \\ 20 \\ 22 \end{bmatrix}$$

We have just computed a composite score of the growth and integration columns (weighting growth twice as heavily as integration).

Matrix-Matrix Multiplication

For an $m \times n$ matrix and an $n \times p$ matrix, the product MW is the $m \times p$ matrix whose (i, k) entry is the dot product of M's i-th row with W's k-th column:

$$(MW)_{ik} = \sum_{j=1}^{n} M_{ij} W_{jk}$$

Each column of MW is exactly what matrix-vector multiplication would have produced using that column of W alone. Multiplying by a matrix is nothing more than multiplying by several vectors, side by side, at once.

Matrix-Matrix Multiplication: Example

In the matrix-vector multiplication example, we applied one weight vector to every row of a matrix. What if want to try *several* weight vectors simultaneously—for example, one composite index that weights growth twice as heavily as integration, and a second that weights them equally?

$$MW = \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 3 & 3 \\ 7 & 9 \\ 6 & 8 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 9 & 5 \\ 9 & 6 \\ 23 & 16 \\ 20 & 14 \\ 22 & 15 \end{bmatrix}$$

Matrix Operations: The Transpose

The transpose of an $m \times n$ matrix M, written $M^\top$, is the $n \times m$ matrix obtained by turning M's rows into columns (equivalently, its columns into rows):

$$(M^\top)_{ij} = M_{ji}$$

Example: $$M = \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 3 & 3 \end{bmatrix}, \qquad M^\top = \begin{bmatrix} 3 & 4 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

M is 3×2, $M^\top$ is 2×3

Note: This is not a simple rotation of the matrix. We are exchanging rows and columns.

Special Classes of Matrices

Square

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

A matrix that has the same number of rows as columns $(m = n)$.

Diagonal

$$D = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

A square matrix whose only nonzero entries lie on the main diagonal (top-left to bottom-right).

Identity

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The diagonal matrix with every $d_i = 1$. Denoted I_n.

Symmetric Matrices

A square matrix S satisfying $S = S^\top$. Its entries are mirrored across the diagonal; $S_{ji} = S_{ij}$ for every i, j.

$$S = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

Note that for any matrix M (square or not), $M^\top M$ is always symmetric:

$$(M^\top M)^\top = M^\top (M^\top)^\top = M^\top M$$